

12/11/2026
Monday

MODULE - I

Introduction to DBMS

Data :- 23

Data stored in db remains available permanently, even after system is shut down or restarted until it is explicitly deleted.

Database :- Collection of interrelated data

① Persistent = permanent storage

② Integrity :- ensures data in db is correct, consistent and not altered

③ Concurrent access ~~concurrent~~ : simultaneous access data

④ Redundancy → due to duplication of data

View of data → see only necessary

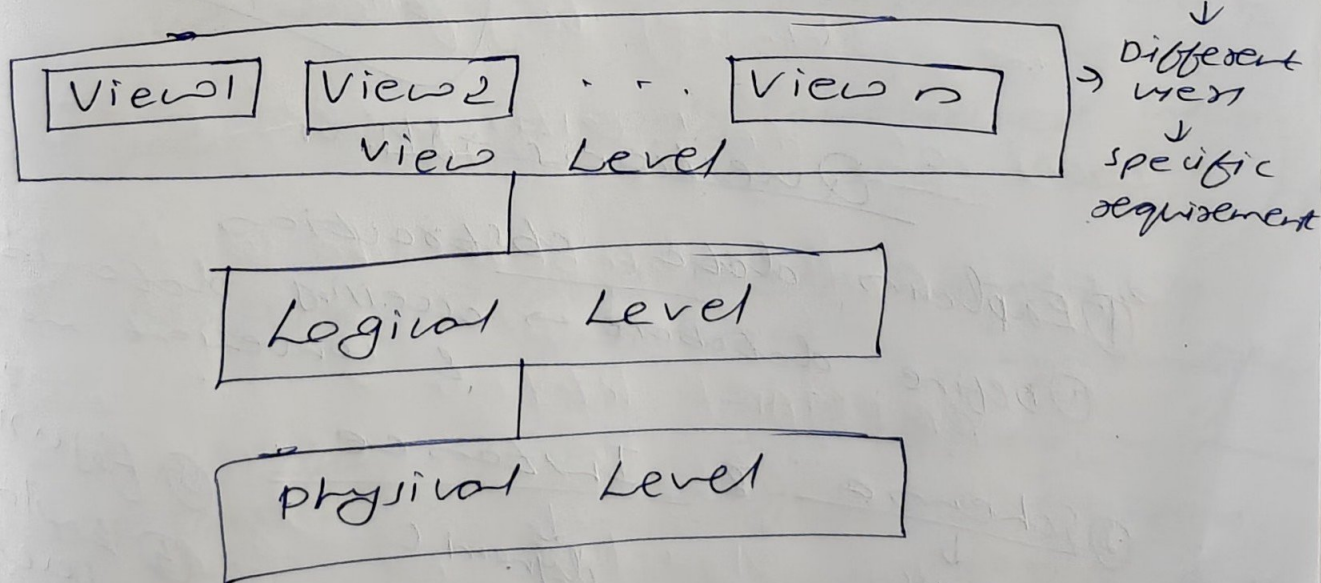
storage, retrieval, updation and deletion manage data in efficient and convenient way.

ES :- Bank System

13/11/26
Tuesday

Data Abstraction in DBMS

only necessary data



mes perspective

→ prophylaxis

→ DBA

specific date,
selection list,
constants

contents

- Abstraction of information

③ Physical Level:- Data storage mechanism

↓
actual database → entire data is stored here

- Backup → recover

Where & how data stored
& accessing mechanism

Questions 98M. or

⑦ Explain data abstraction → accessing a

① Define database → accessing data is efficient & convenient manner.

② Schema : & Instance

↓ ↓

Static Dynamic

③ Figure $\leftarrow$ temp
④ Enplais econ.
level $\leftarrow$ M
lead

② Explain database system Architecture.

view₁, view₂, ... Big different user: -

- Type of user & responsibilities:
- naive user: - no programming
 - sophisticated user: - complete data set view → ~~reference~~
 - Application programmer: - write programs
 - Database Administrator (DBA) → select, update, delete, authorize, privileges.
8M

- Abstraction → 2
- View level → user
- logical level → detailed, DML compiler, DDL
- physical level → database → allocation & indexing

(8M)

③ Advantages of DBMS → 6 points.

④ Limitations of file system

- ⑤ Compare b/w DBMS & file system
- ⑥ Define DBMS → define db schema, manipulation, sharing, security → definition
- ⑦ Avoids Data Redundancy.

⑧ Due to redundancy, inconsistency occur

⑨ Data security → only authorized user access the data

⑩ Integrity → correctness, consistency, consistency. so many conditions

⑪ Conflict resolution → multi-user environment
concurrent access Anomaly avoid conflict
sharing of data

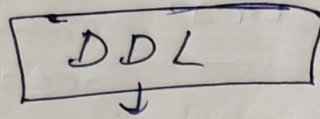
⑫ Data Independence → physical
logical

⑬ Sharing of data

29/11/2025
Thursday

> Data Definition Language (DDL)

- It is used to create the structure of the database.



"Define and manage the structure of the database."

- ① → CREATE
- ② → ALTER
- ③ → DROP
- ④ → TRUNCATE

> Role of DDL

- Defines table, columns, datatypes
- Specific constraints (PRIMARY KEY, FOREIGN KEY)

↓
"Some rules to be followed"

- Does not deal with actual data values.

→ Ex: - Roll numbers should be unique + NOT NULL
↓
PRIMARY KEY

only structure, no data → DDL

Eg :- CREATE TABLE Student(
RollNo INT PRIMARY KEY,

When a DDL statement is files:-

(i) DDL Compiler validates the command
↓
any syntax error

(ii) Table structure is recorded in
data dictionary

(iii) Storage is prepared

(iv) No student data is inserted yet.

Logical level → only structure & data
table & relationships.

② Data Manipulation Language (DML)

DML is used to work with actual
data stored in tables.

"Manipulating data in database"

- ① → INSERT
- ② → UPDATE
- ③ → DELETE
- ④ → SELECT

Role

- ① Add
- ② Modify
- ③ Delete
- ④ Retrieve

DDL → allocate space
DML → insert values
retrieve

→ DML Compiler

- Converts DML statements into low-level instructions
- Works closely with query processor

Ex:- INSERT INTO STUDENT
VALUES (101, 'Pranav', 'MCA');

• DML compiler checks: -

- * Table exists
- * Data-type checks
- * Primary key is unique

DDL is used to only define the structure of the table, its relationships and constraints. It is not concerned with data. In order to add, retrieve, delete or any manipulation of data, we use DML commands.

7 Query Processor

3rd & 4th solutions - The best solution
info.

- ① Accept SQL queries
- ② Decides how to execute them
- ③ Produces correct & optimized results.

DDL command → structure is defined

DDL compiler → metadata stored

DML command

① CREATE

CREATE TABLE MARKS (ROLLNO INT
PRIMARY KEY, COURSE VARCHAR(20),
MARK INT);

② INSERT

INSERT INTO MARKS VALUES (101,
'DBMS', 85);

③ SELECT

SELECT MARK FROM MARKS
WHERE ROLLNO = 101;

SELECT ... FROM ... WHERE
 ↓ ↓ ↓
 DML table condition
 query

PRIMARY KEY constraints
 > Domain constraints
 Entity integrity
 Foreign key constraints

} Type of constraints

• Primary key :- Unique values.

• Foreign key :- A primary key in one table when used in another table become foreign key.

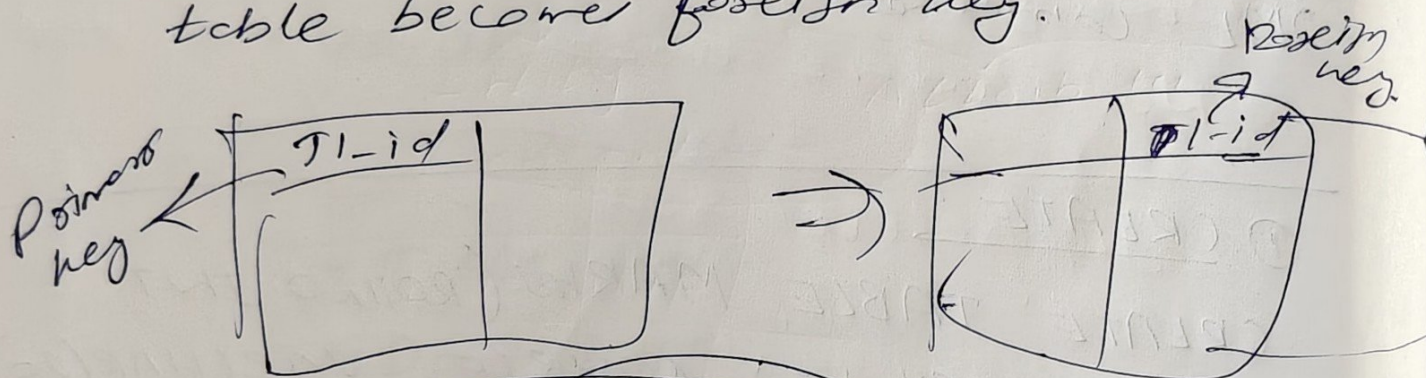


TABLE = RELATION

> What is constraint & constraint violation.

> WHERE clause

specifies the condition while retrieving the value.

> LIKE operator → for comparison / pattern matching

> Range of salary b/w 10000 & 20000

Where salary > 10000 OR salary < 20000

> BETWEEN operator → BETWEEN 1000 AND 2000

> Retrieve data: -

(i) with only 3 locations

(ii) city name is 'Kollam'

Question

> SELECT clause

* FROM → which table to retrieve.

* WHERE → specify the condition by which data to retrieve.

* SELECT ↓

which all columns to select.

(i) column-name

(ii) * → retrieves all columns.

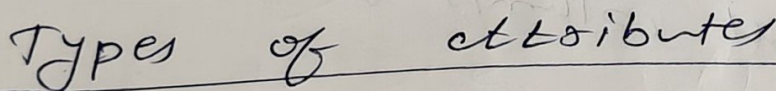
Data Models → overall design

7 ER Model [Entity - Relationship Model]

* Terminologies - real-world objects

② Attributes: - properties about object

ES: - Student → rollno, name, address, phone no.
↓
Entity Attributes



Address

House No. Street State Pincode

Roll no. → Kerala → India

Age calculated from DOB

2003 → 23

③ Single-value attribute :-

Have only one value

④ Multi-value attribute :-

Have more than one value.

ES :- phone numbers

98804765 } More than one
9999 } phone no's

⑤ Stored Attribute :-

ES :- DOB is stored to calculate the age.

① → Single-valued

② → Multi-valued

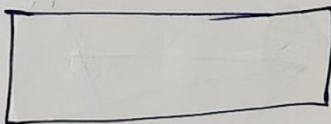
③ → Composite

④ → Derived

⑤ → Stored

7 Symbols

①



Entity

rollno

address

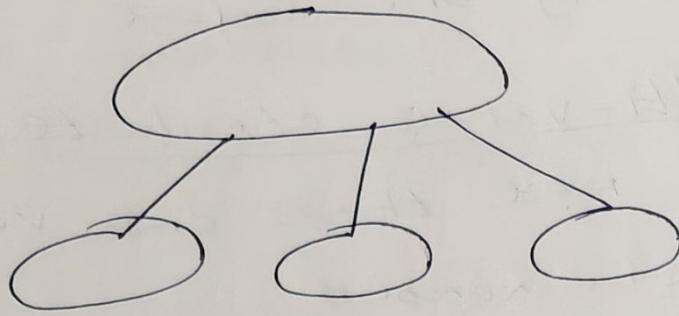
age

age

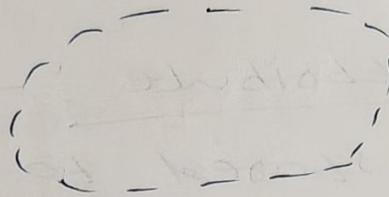
ES :-

Student

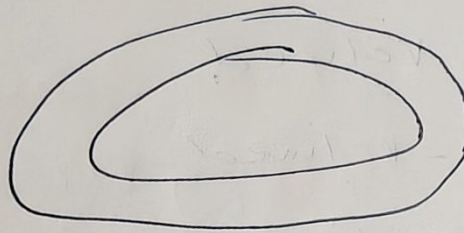
② Composite attribute



③ Derived attribute



④ Multi-valued attribute



⑤ Relationship



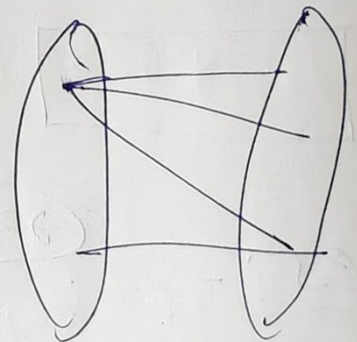
- ~~①~~ ① One - to - One

② One - to - Many

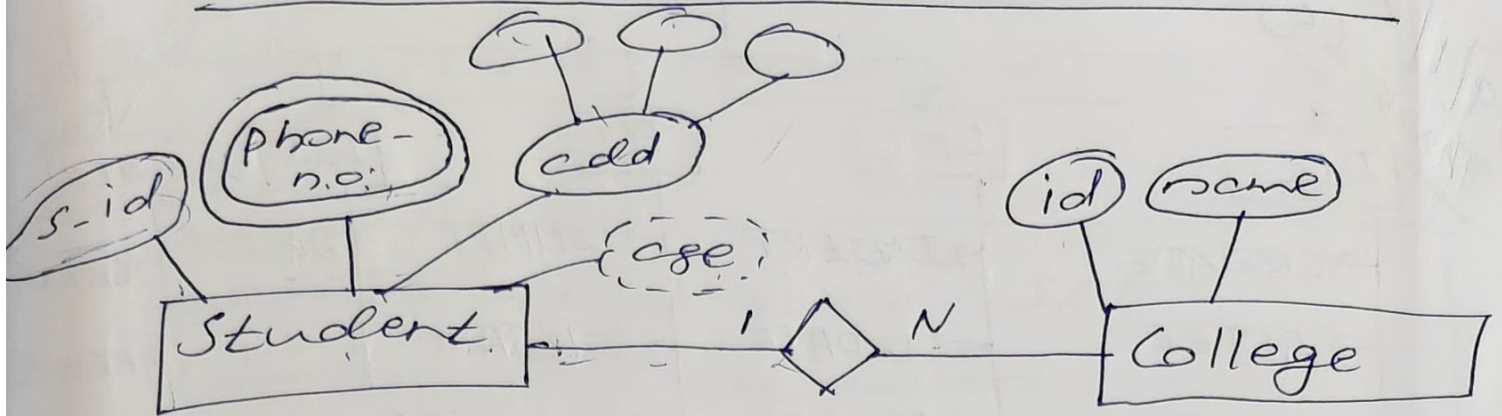
③ Many - to - One

④ Many - to - Many

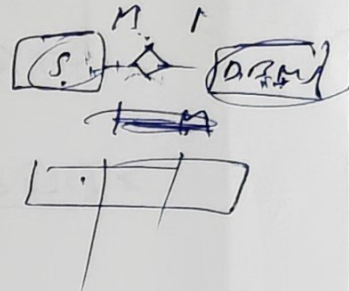
Mapping cardinalities



Example of ER Model



college have
220, 1 to 1
1 to 1, 1
220, 1 to 1
1 to 1, 1 to 1



> 1 - to Many

user - phone no.

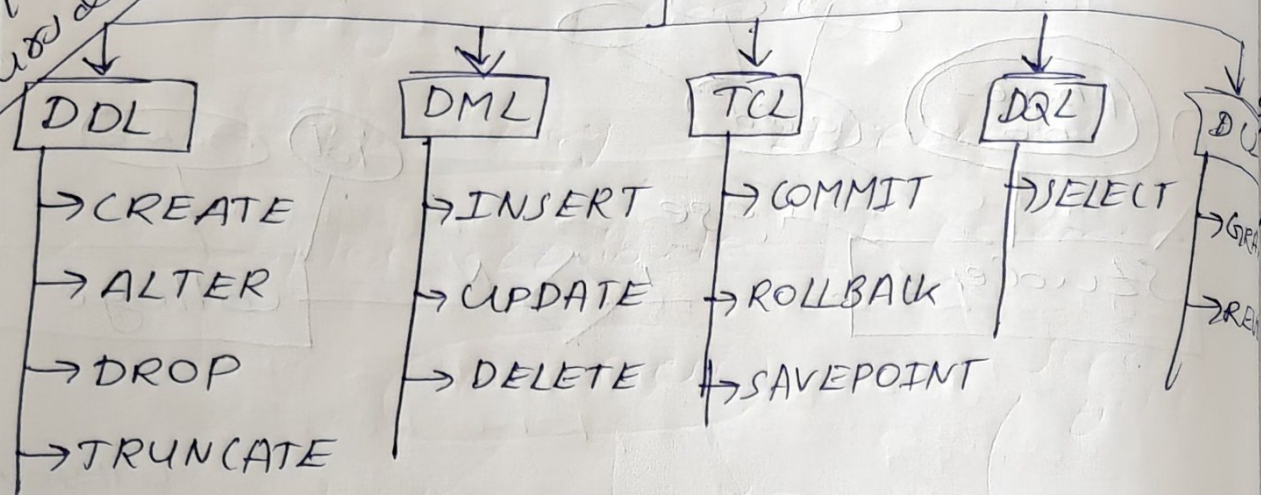
↓
is not an entity

[Attribute, no entity]

> Find examples

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Thursday

SQL Commands



> DCL (Data Control Language)

→ GRANT
→ REVOKE

> DQL

30/11/2025
Friday

> Data Manipulation

① UPDATE Command

SET

```
UPDATE tablename SET  
columnname = expression, ...  
WHERE columnname = expression;
```

* If table delete successfully, returning
of attribute from of table-nya
foreign key relationship, error nrobo.

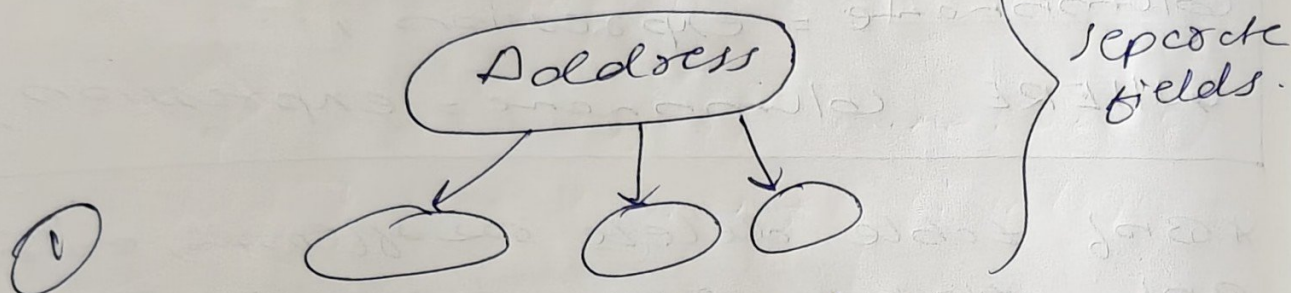
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> ER Model → logical design

• Entity, Relationship

> Foreign key → REFERENCES

> Complex attribute



• In Relational model,
only single-value attribute.

means means separate columns number

↓
But logically ER → info.

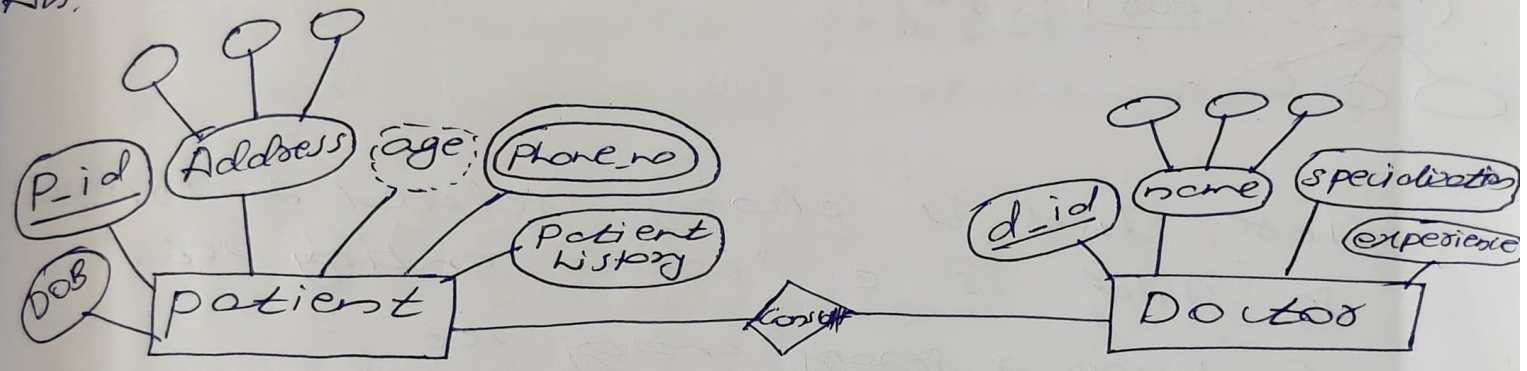
> ER to Relational Model

②

Tutorial

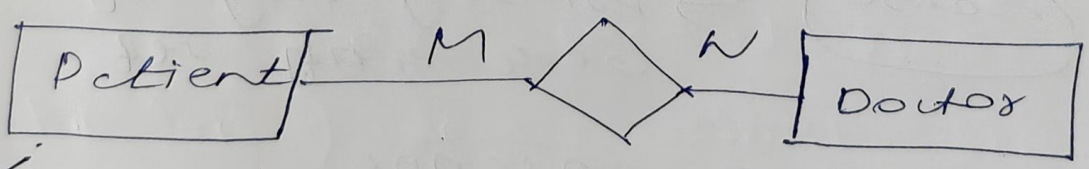
① Construct an ER diagrams for a patient information system. Store patient history and doctors, specialization and experience information. Apply your common sense draw the entities, and the ~~attr~~ relationship and different attribute types (try to include all the attribute types).

Ans:-



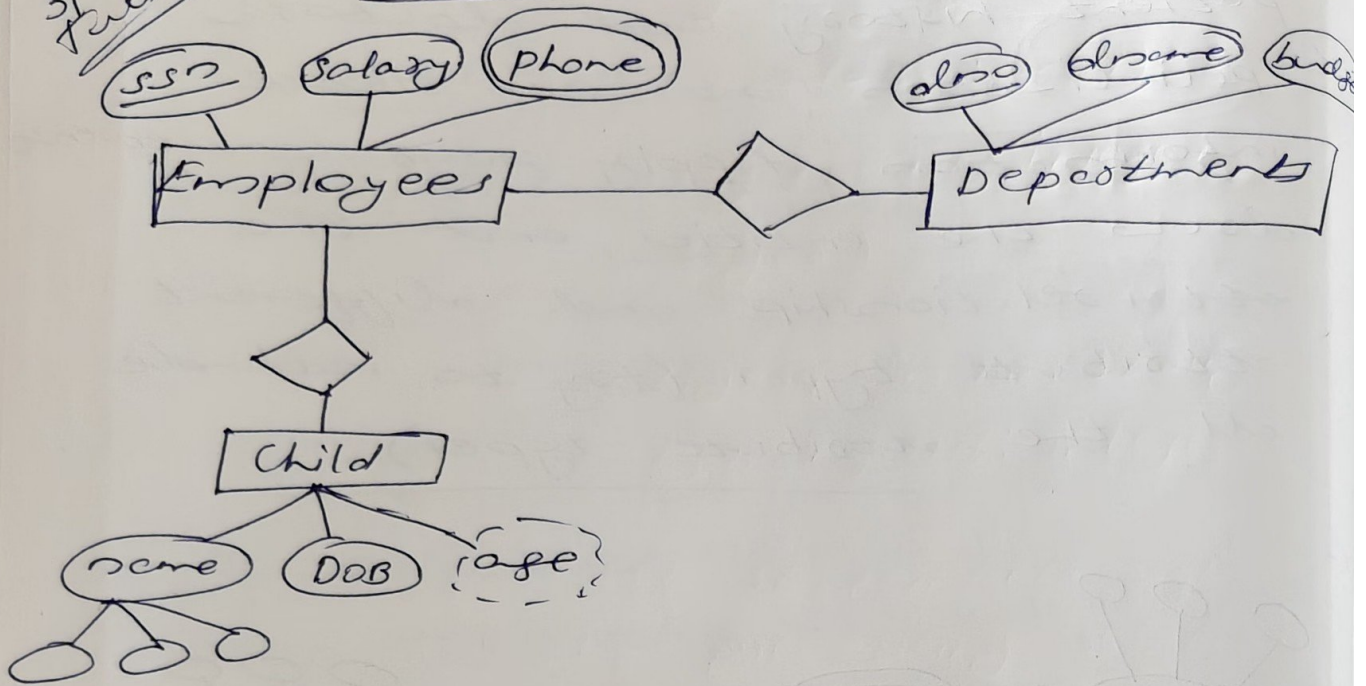
- * System boundary → no 1cb,
- * Availability → specialization, time
- * DOB < Age

H.W



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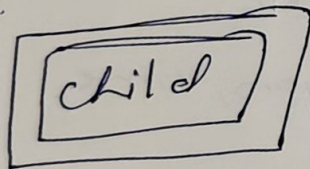
ER Model to Relational Model



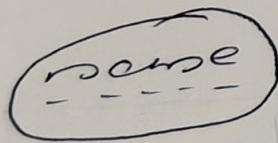
Here, child is always related to Employee
So, child is a "weak entity set."

Employee → strong entity set.

Weak Entity, Relationship



→ weak entity set



→ primary key of weak entity set

ON DELETE CASCADE

Weak Entity set

↓
Whenever Employee is deleted,
the related child needs to be
deleted from the child table.

child table delete on delete cascade
 But employee delete on delete, child
 delete on delete.

Dependents $\rightarrow$ Relationship converted to table.

- ① On delete set null
- ② On delete set default
- ③ On delete cascade

base relation / parent table - the 3rd value
 delete on delete child data delete
 on delete cascade,

On delete cascade

$\rightarrow$ Assertion constraints.

Eno	Ename
A	Project
B	
C	

Dno		
1	MCA	
2	EE	

Work in	
Eno	Dno
A	1
A	2

Foreign
 key

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Problem

- Emp (eid: integer, ename: String, age: integer, salary: real)
- Works (eid: integer, did: integer, pct time: integer)
- Dept (did: integer, dname: String, budget: real, managerid: integer)

Write the SQL statements required to create the preceding relation, including appropriate versions of all primary and foreign key integrity constraints.

Ans:- ① Create Emp table

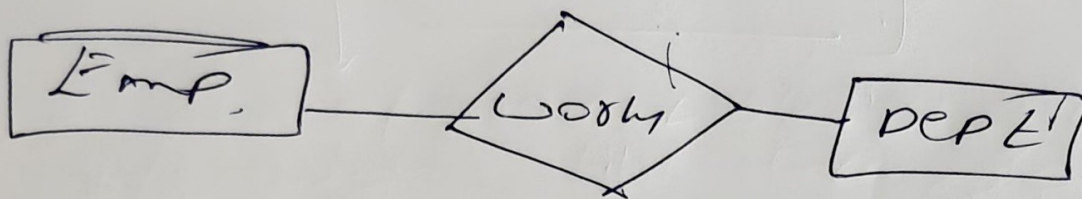
```
CREATE TABLE Emp (eid integer  
PRIMARY KEY, ename VARCHAR(20),  
age integer, salary decimalfloat (20, 2));
```

② Create Works table

```
CREATE TABLE Works (did integer  
PRIMARY KEY, pct time TIME,  
FOREIGN KEY (eid) REFERENCES  
Emp);
```

③ Create Dept table

```
(CREATE TABLE Dept (eid integer  
PRIMARY KEY, ename VARCHAR(20),  
budget NUMERIC(20, 5), FOREIGN KEY  
managerid INTEGER);
```



Add DOW to the emp table

```
INSERT INTO Emp (eid, ename, age,  
salary) VALUES (101, 'DOW', 20, 75,000);
```

Give 10% raise

$$\frac{10}{100} = 0.1$$

```
UPDATE TABLE emp SET salary  
salary * 0.1;
```

Delete

```
DELETE FROM emp where ename =  
"Toy";
```

ON DELETE SET NULL.

9/11
M

~~Def~~

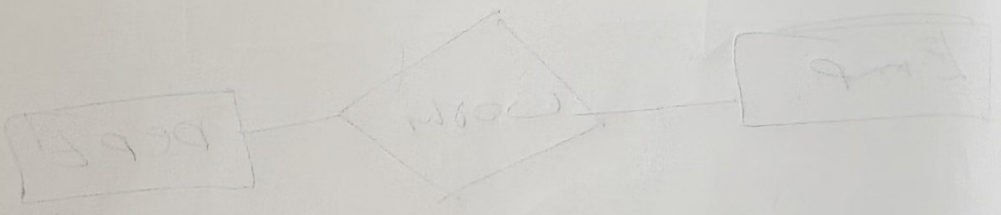
Cardinality $\rightarrow$ NO. of rows
Degree $\rightarrow$ NO. of columns

> Foreign key

Definition + Examples + Tables

• REAL $\rightarrow$ data type

> Views



Ans:-

C

P

a

C

CH

P

FO

E